Exercise 12

On methods of integration

From the school book | Understand | Apply | Higher Order Thinking Skills

First Problems of the integration

1 **Find each of the following :**

(1) $\int x^3 (2x^4 - 7)^5 \, dx$

(2) $\int (x^2 - 4x)^{-3} (x - 2) \, dx$

(3) $\int (1 - x^2) (3x - x^3)^5 \, dx$

(4) $\int x\sqrt{x^2 + 1} \, dx$

(5) $\int (x - 1)\sqrt[3]{x^2 - 2x} \, dx$

(6) $\int \frac{x^2}{(x^3 - 4)^5} \, dx$

(7) $\int \frac{x}{(x^2 + 8)^3} \, dx$

(8) $\int \frac{x^2 + 1}{(x^3 + 3x)^6} \, dx$

(9) $\int \frac{5x}{3x^2 - 1} \, dx$

(10) $\int \frac{12x + 2}{3x^2 + x + 7} \, dx$

(11) $\int \frac{x^3}{\sqrt{2x^4 + 3}} \, dx$

(12) $\int \frac{x^2 + 1}{\sqrt{x^3 + 3x + 5}} \, dx$

(13) $\int \frac{2x - 5}{\sqrt{(x^2 - 5x + 7)^3}} \, dx$

(14) $\int \frac{1}{x^2} \left(1 + \frac{1}{x}\right)^3 dx$

(15) $\int \frac{dx}{\sqrt{x}\left(\sqrt{x} + 2\right)^5}$

(16) $\int \sqrt{\frac{1 + \sqrt{x}}{x}} \, dx$

(17) $\int (2x - 3)^5 (4x + 1)^5 (8x - 5) \, dx$

(18) $\int \sqrt{x^3 - 3x^2 + 3x - 4} \, (x - 1)^2 \, dx$

2 **Find each of the following :**

(1) $\int x (x - 5)^3 \, dx$

(2) $\int x\sqrt{x + 4} \, dx$

(3) $\int x\sqrt{2x + 5} \, dx$

(4) $\int x^2\sqrt{x + 2} \, dx$

(5) $\int (x^2 - 1)\sqrt{x + 1} \, dx$

(6) $\int x^2\sqrt[3]{x - 3} \, dx$

(7) $\int x^3\sqrt{1 - x^2} \, dx$

(8) $\int x^3\sqrt[5]{(x^2 - 1)^2} \, dx$

(9) $\int x^3 (x^2 - 1)^5 \, dx$

(10) *(1st Session 2017)* $\int \frac{x}{x + 1} \, dx$

(11) $\int \frac{x^2}{x-1}\,dx$

(12) $\int \frac{x}{(x+1)^2}\,dx$

(13) $\int \frac{x}{\sqrt{2x+2}}\,dx$

(14) $\int \frac{x+1}{\sqrt{x-1}}\,dx$

(15) $\int \frac{x\,dx}{\sqrt{4x-1}}$

(16) $\int \frac{\sqrt{x}\,dx}{(\sqrt{x}-1)^6}$

(17) $\int \frac{x^2}{\sqrt{2x-1}}\,dx$

(18) $\int \frac{dx}{\sqrt[3]{\sqrt{x}+1}}$

(19) $\int \frac{x^3\,dx}{\sqrt{(x^2+3)^3}}$

(20) $\int x^{15}\left(\frac{1}{x}+\frac{1}{x^2}\right)^7 dx$

(21) $\int 9x^3\sqrt[3]{\frac{4}{x^3}-3}\,dx$

(22) $\int \frac{2x-1}{(2x+3)^2}\,dx$

(23) $\int \frac{(x+3)^3-27}{x}\,dx$

(24) $\int x\left(\sqrt{x}+2\right)^6 dx$

(25) $\int \sqrt{x}\left(\sqrt{x}-3\right)^4 dx$

(26) $\int \sqrt{x^{\frac{3}{2}}+x}\,dx$

(27) $\int x \sin x^2\,dx$

(28) $\int x \tan 3x^2\,dx$

3 Choose the correct answer from the given ones :

(1) $\int x^n\,dx = \frac{x^{n+1}}{n+1} + c$ such that $n \neq$ .....

(a) zero (b) 1 (c) 2 (d) – 1

(2) *(Speciman 2025)* Use the substitution $z = \sqrt{x}$, then the integration $\int \frac{e^{\sqrt{x}}}{\sqrt{x}}\,dx$ will be in form of .....

(a) $\int 4e^z\,dz$ (b) $\int 2e^z\,dz$ (c) $\int \frac{1}{2}e^z\,dz$ (d) $\int \frac{1}{4}e^z\,dz$

(3) *(Speciman 2025)* Use the substitution $z = \sqrt{x}$, then the integration $\int \frac{\ln\sqrt{x}}{2\sqrt{x}}\,dx$ will be in form of .....

(a) $\int \ln z\,dz$ (b) $\int 2\ln z\,dz$ (c) $\int 4\ln z\,dz$ (d) $\int \frac{1}{2}\ln z\,dz$

(4) $\int 6x^5\left(1-\frac{1}{x}\right)^5 dx =$ ..... + c

(a) $6(x-1)^6$ (b) $(x-1)^6$ (c) $\frac{1}{6}(x-1)^6$ (d) $(6x-6)^6$

(5) *(Speciman 2025)* $\int 12\,x^2\,(5-4\,x^3)^{-5}\,dx =$ ..... $+ c$

(a) $\frac{-1}{4}(5-4\,x^3)^{-4}$ (b) $\frac{1}{4}(5-4\,x^3)^{-4}$

(c) $\frac{-1}{6}(5-4\,x^3)^{-6}$ (d) $\frac{1}{6}(5-4\,x^3)^{-6}$

(6) $\int \frac{(4\,x^2-4\,x+1)^7}{(2\,x-1)^2}\,dx =$ ..... $+ c$

(a) $\frac{1}{13}(2\,x-1)^{13}$ (b) $\frac{1}{26}(2\,x-1)^{13}$ (c) $\frac{2}{3}(2\,x-1)^{13}$ (d) $\frac{1}{16}(2\,x-1)^8$

(7) If $\int 3\,x^2\,(x^n+1)^5\,dx = \frac{(x^n+1)^6}{6} + c$, then n = ...

(a) 2 (b) 3 (c) 5 (d) 6

(8) *(Speciman 2025)* If $\int x^2\,e^x\,dx = (a\,x^2+b\,x+d)\,e^x + c$, then $a+b+d =$ .....

(a) -2 (b) -1 (c) zero (d) 1

(9) *(Speciman 2025)* If $\int x\cos x\,dx = a\,x\sin x + b\cos x + c$, then $a+b =$ .....

(a) -2 (b) -1 (c) 1 (d) 2

(10) If $\int \frac{x^2\,dx}{\sqrt{2\,x^3+1}} = n\sqrt{2\,x^3+1} + c$, then n = ...

(a) 3 (b) $\frac{1}{3}$ (c) 6 (d) $\frac{1}{6}$

(11) $\int \frac{d}{dx}\left(x^5+\sqrt{x}\right)dx =$ .....

(a) $\frac{1}{6}x^6 + \frac{1}{2\sqrt{x}} + c$ (b) $x^4 + \frac{1}{2}x^{-\frac{1}{2}}$

(c) $x^5+\sqrt{x}$ (d) $x^5 + \sqrt{x} + c$

(12) If $f(x) = \int \frac{1}{x}\,dx$, then $f'(2) =$ .....

(a) not exist. (b) $\frac{1}{x} + c$ (c) 2 (d) $\frac{1}{2}$

(13) $\int x\,(x^2+3)^5\,dx =$ ..... $+ c$

(a) $\frac{1}{6}(x^3+3)^6$ (b) $\frac{1}{12}(x^2+3)^6$ (c) $\frac{1}{4}(x^2+3)^4$ (d) $\frac{1}{8}(x^2+3)^4$

(14) *(Speciman 2025)* $\int \frac{\ln 2x}{x}\,dx =$ ..... $+ c$

(a) $\frac{1}{2}(\ln 2x)^2$ (b) $\ln|\ln 2x|$

(c) $\frac{1}{2}x^2 \ln 2x - \frac{1}{4}x$ (d) $2(\ln 2x)^2$

(15) $\int x \cdot f'(x^2) \cdot f(x^2)\, dx =$ ..... $+ c$

(a) $\frac{1}{4}[f(x^2)]^2$ (b) $\frac{1}{2}[f(x^2)]^2$ (c) $[f(x^2)]^2$ (d) $2[f(x^2)]^2$

(16) $\int x f'(1-x^2)\, dx =$ ..... $+ c$

(a) $-f(1-x^2)$ (b) $-2f(1-x^2)$ (c) $-\frac{1}{2}f(1-x^2)$ (d) $x f(1-x^2)$

(17) $\int \left[\lim_{h \to 0} \frac{f(x+h)-f(x)}{h}\right] dx =$ .....

(a) $f'(x)$ (b) $f(x) + c$ (c) $f'(x) + c$ (d) $f''(x) + c$

(18) If y, z are functions of x, then $\int y\, dz + \int z\, dy =$ .....

(a) $d(yz)$ (b) $\int yz\, dx$ (c) $yz + c$ (d) $y + z + c$

(19) If $\int (2x+3) \ln x\, dx = yz - \int z\, dy$, then $yz =$ .....

(a) $2x \ln x$ (b) $(2x+3) \ln x$

(c) $\frac{1}{2}(2x+3) \ln x$ (d) $x(x+3) \ln x$

(20) If $\int (2x-1) e^{2x+3}\, dx = yz - \int z\, dy$, then $\int z\, dy =$ ..... $+ c$

(a) e^{2x+3} (b) $\frac{1}{2}e^{2x+3}$ (c) $-e^{2x+3}$ (d) $-\frac{1}{2}e^{2x+3}$

(21) *(Speciman 2025)* $\int \sqrt{1+\sqrt{x}}\, dx =$ ..... $+ c$

(a) $\frac{2}{5}(1+\sqrt{x})^{\frac{5}{2}} + \frac{1}{2}(1+\sqrt{x})^{\frac{3}{2}}$ (b) $\frac{4}{5}(1+\sqrt{x})^{\frac{5}{2}} - \frac{4}{3}(1+\sqrt{x})^{\frac{3}{2}}$

(c) $\frac{4}{5}(1+\sqrt{x})^{\frac{5}{2}} + \frac{4}{3}(1+\sqrt{x})^{\frac{3}{2}}$ (d) $\frac{4}{5}(1+\sqrt{x})^{\frac{5}{2}} + \frac{4}{3}(1+\sqrt{x})^{\frac{1}{2}}$

(22) $\int (3x+2) \sin x\, dx =$ ..... $+ c$

(a) $(3x+2) \cos x + 3 \sin x$ (b) $-(3x+2) \cos x + 3 \sin x$

(c) $(3x+1) \cos x + 2 \sin x$ (d) $-(x+1) \cos x - 3 \sin x$

(23) *(2nd Session 2023)* $\int \ln \frac{1}{x}\, dx =$ ..... $+ c$

(a) $x[-1 + \ln x]$ (b) $x[1 - \ln x]$ (c) $x[1 + \ln x]$ (d) $x[-1 - \ln x]$

(24) $\int e^x (1 + \tan x + \tan^2 x) =$ ..... $+ c$

(a) $e^x \tan x$ (b) $e^x \sec x$ (c) $e^x \sin x$ (d) $e^x \cos x$

(25) $3 \int \sin 2x \sin^4 x\, dx =$ ..... $+ c$

(a) $\sin^4 x \cos^2 x$ (b) $\cos^4 x \sin^2 x$

(c) $\sin^6 x$ (d) $\cos^6 x$

(26) *(2nd Session 2021)* If $z = x + 2$, then $\int \frac{x-2}{x^2+4x+4}\, dx =$ ..... $+ c$ where c is constant.

(a) $\ln|z| - \frac{4}{z}$ (b) $\ln|z| + \frac{4}{z}$ (c) $\frac{-1}{2z^2} - \frac{4}{3z^3}$ (d) $\frac{-1}{2z^2} + \frac{2}{3z^3}$

(27) *(1st Session 2025)* $\int e^{\sin x} \times \sin 2x \, dx = \ldots\ldots\ldots\ldots + c$

(a) $e^{\sin x}(\sin x - 1)$ (b) $2e^{\sin x}(1 - \sin x)$

(c) $2e^{\sin x}(\sin x - 1)$ (d) $e^{\sin x}(1 - \sin x)$

(28) *(Speciman 2025)* $\int x \ln x \, dx = \ldots\ldots\ldots\ldots + c$

(a) $\frac{1}{4}x^2(\ln x^2 - 1)$ (b) $\frac{1}{4}x^2 \ln x - 1$

(c) $\frac{1}{4}x^2 \ln x + 1$ (d) $\frac{1}{2}x^2 \ln x + \frac{1}{4}$

(29) *(Trial 2021)* $\int \frac{9}{2}\sqrt{x} \ln x \, dx = \ldots\ldots\ldots\ldots + c$

(a) $x^{\frac{3}{2}}(\ln x^3 - 2)$ (b) $x^{\frac{3}{2}}(\ln x - 2)$ (c) $x^{\frac{3}{2}}(\ln x^3 + 2)$ (d) $x^{\frac{3}{2}}(\ln x + 2)$

(30) *(Trial 2023)* $\int \ln\sqrt{x} \, dx = \ldots\ldots\ldots\ldots + c$ where c is constant.

(a) $\frac{1}{2}x \ln\frac{e}{x}$ (b) $\frac{1}{2}x\frac{\log x}{\log e}$ (c) $\frac{1}{2}x \ln\frac{x}{e}$ (d) $\frac{1}{2}x\frac{\log e}{\log x}$

4 Find each of the following :

(1) $\int x e^x \, dx$

(2) $\int 4x e^{2x} \, dx$

(3) *(2nd session 2017)* $\int \frac{x}{e^{2x}} \, dx$

(4) $\int \frac{3x + 5}{e^{2x}} \, dx$

(5) $\int \ln x \, dx$

(6) $\int \ln(x + 1) \, dx$

(7) $\int \ln\left(\frac{x}{2}\right) dx$

(8) $\int \ln x^2 \, dx$

(9) $\int x \ln(x + 1) \, dx$

(10) $\int x^3 \ln x \, dx$

(11) $\int \frac{1}{x \ln x} \, dx$

(12) $\int \frac{1}{x(\ln x)^3} \, dx$

(13) $\int \sqrt{x} \ln x \, dx$

(14) $\int \frac{\ln x}{\sqrt{x}} \, dx$

(15) $\int \ln(5x) \frac{dx}{x}$

(16) $\int (\ln x)^3 \frac{dx}{x}$

(17) $\int \ln x \frac{dx}{x^2}$

(18) $\int x^2 \ln x^2 \, dx$

(19) $\int e^x(\cot x - \csc^2 x) \, dx$

(20) $\int e^x \sec x(1 + \tan x) \, dx$

(21) $\int e^x \csc x(1 - \cot x) \, dx$

(22) $\int x \sin x \, dx$

(23) $\int x \cos 4x \, dx$

(24) $\int x \sec^2 x \, dx$

5 **Find each of the following :**

(1) $\int 2x\, e^{3x^2+1}\, dx$

(2) $\int \frac{e^{2x}}{e^{2x}+3}\, dx$

(3) $\int \frac{4\, e^{3x}}{1-e^{3x}}\, dx$

(4) $\int (3-x)\, e^{6x-x^2}\, dx$

(5) $\int \frac{e^{\sqrt{x+1}}}{\sqrt{x+1}}\, dx$

(6) $\int \frac{e^{-x}-1}{e^{-x}+x}\, dx$

(7) $\int \frac{e^{2x}+x}{e^{2x}+x^2}\, dx$

(8) $\int \frac{x^2+1}{x^3-x}\, dx$

6 **Find each of the following :**

(1) $\int 9x^2 e^{3x}\, dx$

(2) $\int x^2 e^{-3x}\, dx$

(3) $\int x^3 e^{x^2}\, dx$

(4) $\int x^2 e^{x+2}\, dx$

(5) $\int (x+1)^2 e^{2x}\, dx$

(6) $\int (x e^x)^2\, dx$

(7) $\int \frac{x e^x}{(x+1)^2}\, dx$

(8) $\int \frac{x^3 e^{x^2}}{(x^2+1)^2}\, dx$

(9) $\int \frac{e^{\frac{1}{x}}}{x^2}\, dx$

(10) $\int (\ln x)^2\, dx$

(11) $\int x (\ln x)^2\, dx$

(12) $\int x^3 e^x\, dx$

7 **Find by two different methods each of the following :**

(1) $\int x\sqrt{4-x}\, dx$

(2) $\int \frac{x}{\sqrt{4+5x}}\, dx$

(3) $\int \frac{4x}{\sqrt[3]{2x+1}}\, dx$

Second Problems on applications of integration

1 **Choose the correct answer from those given :**

(1) The equation of the curve whose y-intercepted part is 7 units from the negative y-axis and the slope of its tangent at any point is $3x^2 + 2$ is .....

(a) $y = x^3$ (b) $y = x^3 - 2x$ (c) $y = x^3 - 7$ (d) $y = x^3 + 2x - 7$

(2) The slope of the perpendicular to the curve at any point on it (x , y) equals $3 - 2x$, then the equation of the curve given it passes through the point $(1 , 1)$ is .....

(a) $y = \frac{1}{2} \ln|2x - 3| + 2$ (b) $y = \frac{1}{2} \ln|2x - 3| + 1$

(c) $y = \ln|2x - 3| + 4$ (d) $y = \ln|2x - 3| + 1$

(3) An empty container capacity is 1400 cm^3 , water is poured in it at rate $(2t + 50)$ cm^3 /sec. where (t) is the time in seconds , then the needed time to fill the container = ... sec.

(a) 10 (b) 15 (c) 20 (d) 30

(4) *(2nd Session 2024)* If the curve of the function f passes through the origin point and $\grave{f}(x) = \frac{1}{2}x^2 + 3x$, then the inflection point of the curve of the function f is .....

(a) $(-3 , 12)$ (b) $(-6 , 18)$ (c) $(0 , 0)$ (d) $(-3 , 9)$

(5) If $f'(x) = \ln x$ and $f(1) = 1$, then $f(2)$ = ...

(a) $\ln\sqrt{2}$ (b) $\ln 2$ (c) $\ln 4$ (d) $\ln 8$

(6) If slope of the tangent to curve of the function $y = f(x)$ at $x = 1$ equals $-\frac{1}{2}$ and $f''(x) = x + 1$, then $f'(2)$ = ...

(a) -2 (b) -1 (c) 1 (d) 2

(7) *(2nd Session 2022)* If the slope of the tangent to the curve of the function $y = f(x)$ at any point (x , y) which lies on it equals $(4x^3 - 12x^2)$ and the function f has a local minimum value at the point $(3 , -27)$, then $f(1)$ = ...

(a) -8 (b) 3 (c) 8 (d) -3

(8) *(Speciman 2025)* If the slope of the tangent to the curve at any point (x , y) on it is $\frac{\ln x}{x}$, the curve passes through the point $(e^2 , 3)$, then the equation of the curve is .....

(a) $y = \frac{1}{2}\ln x + 2$ (b) $y = \ln x + 1$ (c) $y = 2\ln x - 1$ (d) $y = \frac{1}{2}(\ln x)^2 + 1$

(9) If $f''(x) = 6x - 4$, $f'(1) = 2$, $f(0) = -4$, then $f(x)$ = ...

(a) $-2x^2 + 3x - 4$ (b) $x^3 - 2x^2 + 3x$

(c) $3x^2 - 4x - 4$ (d) $x^3 - 2x^2 + 3x - 4$

(10) If the slope of the tangent to the curve : $y = f(x)$ at any point on it equals $6x + a$ where a is constant and the equation of the tangent to the curve at the point $(1 , -1)$ is $y = 4 - 5x$, then the equation of the curve is .....

(a) $y = 3x^2 - 4$ (b) $y = 3x^2 - 11x + 7$

(c) $y = 3x^2 - 4x + 7$ (d) $y = 3x^2 - 11x - 4$

(11) If the slope of the tangent to the curve of the function $y = f(x)$ at any point on it equals $\frac{1}{x\sqrt{3+\ln x}}$ where $x > 0$ and the curve passes through $(e , 4)$, then the relation between x , y is .....

(a) $y = \sqrt{3+\ln x}$ (b) $y = 2\sqrt{3+\ln x}$

(c) $y = 3 + \ln x$ (d) $y = 2\sqrt{3+\ln x} - 5$

(12) *(2nd Session 2021)* If $3y^2 \frac{dy}{dx} = 2x + 1$ and $x = 1$ when $y = -1$, then the relation between x , y is .....

(a) $y^3 = x^2 + x - 3$ (b) $y^3 = x^2 - 2$

(c) $y^3 + x^2 = 0$ (d) $y^3 + 1 = x^2 - x$

(13) *(2nd Session 2021)* If $y = f(x)$, $\frac{d^2x}{dy^2} = e^{y+1}$ and the tangent to the curve of the function f at the point $(x , -1)$ which lies on it is parallel to the straight line : $y = x - 3$, then the slope of the tangent to the curve of the function f at any point (x , y) lies on it equals .....

(a) e^{y+1} (b) e^{2y+2} (c) e^{-y-1} (d) e^{-2y-2}

(14) *(Speciman 2025)* If the slope of the tangent of the curve $y = f(x)$ is given by the relation $\frac{dy}{dx} = x e^{x}$ and the curve passes through the point $(0 , 4)$, then $f(1)$ = ...

(a) 5 (b) 4 (c) 3 (d) 2

2 Find the equation of the curve passing through the point A $(2 , 3)$, and the slope of the normal on it at any point (x , y) is $3 - x$ «$y = \ln|x - 3| + 3$»

3 If the slope of a tangent to a curve at a point (x , y) lying on it is $x\sqrt{x+1}$, find the equation of the curve known that the curve passes through the point $\left(0 , \frac{11}{15}\right)$

«$y = \frac{2}{5}(x+1)^{\frac{5}{2}} - \frac{2}{3}(x+1)^{\frac{3}{2}} + 1$»

4 Find the equation of the curve passing through the point $(0 , 1)$ and its slope of tangent at the point (x , y) lying on it equals $x\sqrt{x^2+1}$ «$y = \frac{1}{3}(x^2+1)^{\frac{3}{2}} + \frac{2}{3}$»

5 The slope of the normal to the curve at any point on it (x , y) is $\frac{\sqrt{x}}{x-1}$ where $x \in \mathbb{R}^+ - \{1\}$, find the curve's equation , given that it passes by the point $(9 , -12)$ «$y = -\frac{2}{3}x\sqrt{x} + 2\sqrt{x}$»

6 The slope of the normal to the curve at any point on it (x, y) equals $\frac{-2y}{x}$, find the curve equation, given that it intercepts 3 units from the positive part of y-axis. « $2y^2 = x^2 + 18$ »

7 The slope of the tangent to the curve of the function f at any point on it (x, y) is given by the relation $g(x) = x e^{3x}$ find the curve equation given that it passes through $(\frac{1}{3}, 5)$

« $y = \frac{1}{3} x e^{3x} - \frac{1}{9} e^{3x} + 5$ »

8 The slope of the normal to a curve is $-(\ln x)^{-2}$ find the curve's equation given that it passes through (e, e) « $y = x(\ln x)^2 - 2x \ln x + 2x$ »

9 If the slope of the tangent to the curve of the function f at any point (x, y) lying on it is given by the relation $g(x) = \frac{x e^x}{(x+1)^2}$, find the equation of the curve if it passes through the point $(1, 2e)$

« $y = \frac{-x e^x}{x+1} + e^x + \frac{3}{2} e$ »

10 ***(2nd Session 2006)*** A liquid is leaking from a small hole in the bottom of a vessel filled with the liquid. If the volume of the liquid changes at the rate of $(0.4t - 40)$ cm^3/sec, where t represents the time in seconds and the volume of the liquid is 980 cm^3 after 30 seconds from the start of leaking. Find the capacity of the vessel. When does the vessel become empty ? « 2000 cm^3, 100 sec. »

11 ***(1st Session 2002)*** If the slope of the tangent to the curve $y = f(x)$, at any point (x, y) on it is given by the relation $\frac{dy}{dx} = \frac{x}{y}$, find the equation of the curve, given that it passes through the point $(1, 2)$ « $y^2 = x^2 + 3$ »

12 ***(1st Session 2020)*** Find the equation of the curve passes through the point $(1, 0)$ and the slope of its tangent at any point on it equals $x e^y$ « $e^{-y} + \frac{1}{2} x^2 = 1\frac{1}{2}$ »

13 ***(1st Session 2006)*** The slope of the tangent at any point (x, y) on the curve $y = f(x)$ is equal to $3x^2 - 6x - 9$ and the local maximum of the function f is 17, find the local minimum of the function f « – 15 »

14 Find the equation of the curve $y = f(x)$ if $\frac{d^2y}{dx^2} = ax + b$ where a and b are two constants and the curve has an inflection point at the point $(0, 2)$ and a local minimum value at the point $(1, 0)$, then find the local maximum value to this curve.

« $y = x^3 - 3x + 2$, 4 »

15 ***(2nd Session 2007)*** If the rate of change of the slope of the tangent to a curve at any point on it is equal to $6x - 2$ and the slope of the tangent at the point $(3, 1)$ that lies on the curve equals 2 Find the equation of this curve. « $y = x^3 - x^2 - 19x + 40$ »

16 *(1st Session 2008)* Find the equation of the curve $y = f(x)$, such that $\frac{d^2 y}{dx^2} = \frac{2}{x^3}$ and the equation of its tangent at the point $(2 , \frac{5}{2})$ which lies on the curve is : $3x - 4y + 4 = 0$

« $y = \frac{1}{x} + x$ »

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17 *(1st Session 2013)* If : $f''(x) = 6x - 4$ at any point on the curve of the function $y = f(x)$ and the curve passes through the point $(1 , -5)$ and the function has a local minimum value at $x = 3$, find the equation of this curve and the local maximum value of the function.

« $f(x) = x^3 - 2x^2 - 15x + 11$, $\frac{697}{27}$ »

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18 If the slope of the tangent to the curve $y = f(x)$ at any point on it equals $6x^2 + bx$ and $f(0) = 5$, $f(2) = -3$, find the value of the constant b , then sketch the curve of the function f

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19 If the slope of the tangent to the curve of the function f at any point (x , y) on the curve is $3(x^2 - 1)$, find the local maximum and minimum values to the curve of the function f and the inflection points if existed known that the curve passes through the point $(-2 , -1)$, then sketch this curve.

« 3 , –1 , (0 , 1) »

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20 If the rate of change of the slope of the tangent to a curve at any point (x , y) on it is $6(1 - 2x)$ and the curve has a critical point when $x = 1$ and the function has a local minimum value equals 4

(1) Find the equation of the normal to the curve when $x = -1$

(2) Sketch the curve of the function and show the maximum and minimum values and the inflection points if existed.

« $x - 12y + 109 = 0$ »

Higher skills

21 **Find the integral of each of the following :**

(1) $\int e^x \sin x \, dx$

(2) $\int \frac{x}{\sin^2 5x} \, dx$

(3) $\int \sec^3 x \, dx$

(4) $\int \cos(\ln x) \, dx$